Victor Petrov

650-382-7248 | 799vpetrov@gmail.com | linkedin.com/in/vpetrov-codes | v7petrov.github.io

Education

California Polytechnic University, San Luis Obispo San Luis Obispo, CA
Bachelor of Science in Computer Engineering, ***GPA 3.7/4.0*** *June 2026*

Relevant Coursework: Data Structures & Algorithms, Real Time Operating Systems, Computer Networking, Computer Security, Modern Computer Architecture, Digital Signals & Systems, Classical Controls

Experience

Controls Software Engineer Intern June 2025 – Sep 2025
General Atomics *Poway, CA*

- Maintained and debugged a large-scale C++ flight simulator codebase (200K+ lines), identifying and resolving critical issues across multiple interconnected systems with real-time networking requirements
- Overhauled the build system by reengineering CMake configuration files, reducing build times by 45% and improving cross-platform compilation reliability
- Developed Python and Bash automation scripts to streamline development workflows, eliminating 15+ hours of manual build and deployment work per week
- Optimized embedded OpenCV and ONNX Runtime implementations through hardware-level profiling and code tuning, achieving 300% faster inference speeds and reducing frame processing latency by 65%
- Led cross-platform migration effort, porting a complex Windows-based flight simulator (150+ source files) to Linux, addressing OS-specific dependencies and system calls while maintaining feature parity
- Implemented and debugged network communication protocols across 8+ interconnected processes, ensuring sub-10ms latency and 99.9% reliability for inter-process communication over TCP/UDP sockets in a distributed real-time system

ISA Student Assistant Sep 2024 – Present
California Polytechnic University *San Luis Obispo, CA*

- Served as Teaching Assistant for Digital Design course, instructing 30+ students on hardware description languages (Verilog, SystemVerilog) and computer architecture fundamentals
- Conducted weekly lab sessions providing hands-on guidance for FPGA-based projects and debugging student designs
- Mentored students on debugging techniques for hardware simulation tools (ModelSim/Vivado), waveform analysis, and testbench development

Embedded Software Intern June 2024 – Sep 2024
LEGUAN MicroElectronics Co. *Shangdong, China*

- Developed AI-enhanced control systems for industrial crystal growth furnaces using GMDH algorithms, optimizing real-time PLC operations
- Built fault-tolerant communication layer between AI inference system and industrial PLCs via standard networking protocols

Projects

Real-Time Motor Control System | *C/C++, Embedded Linux, Bare-metal Firmware*

- Debugged and adapted open-source motor controller firmware written in C for custom PCB hardware, resolving initialization failures, PWM signal generation issues, and communication protocol mismatches
- Created custom terminal commands to interface and improve debugging process, speeding up loading times by 10x
- Calibrated motor controller parameters based on telemetry data analysis, optimizing performance while maintaining safety margins
- Validated diagnostic and error handling routines in firmware, testing robust fault detection for overcurrent, overvoltage, and overtemperature conditions with safe-state fallback behavior

Full Stack 4-Axis Robot Arm | *PCB Design, Bare Metal C, MicroPython, I2C, BLE*

- Designed and brought up compact motor controller PCBs using RP2040, enabling distributed control architecture with I2C communication between 4 nodes
- Implemented inverse kinematics engine in MicroPython and optimized to bare metal C for real-time motor control
- Developed BLE-enabled web application interface for wireless robot control and coordination

GridGuard - Residential Battery UPS | *Customer Development, Battery Chemistry, Power Electronics*

- Led full product development cycle from customer discovery to working prototype of home power supply
- Engineered battery testing and validation protocol to safely repurpose used li-ion cells, reducing UPS cost by 50%
- Designed power management circuitry to convert repurposed battery packs into grid-backup power system for residential customers

Function Generator and Oscilloscope | *C++, I2C, PWM, UART*

- Programmed function generator and oscilloscope on STM32 without using hardware abstraction layer (HAL), debugging through JTAG hardware
- Implemented I2C, UART, SPI communication protocols through direct register-level programming using STM32 documentation

Real-Time Telemetry Platform | *Python, Flask, React, Docker*

- Built scalable full-stack web application with React frontend and Flask backend for real-time vehicle telemetry
- Implemented comprehensive test suite including unit tests, integration tests, and end-to-end automated testing
- Designed robust error handling and logging systems to troubleshoot and resolve application defects

Real-time Control System for Pipetting Robot | *C++, Linux, Embedded Systems*

- Developed embedded software for pipetting robot with 3-axis frame control using C++ on Linux-based system
- Implemented network communication protocols to control entire machine via web server interface and remote commands

Underwater Remote Operating Vehicle (ROV) | *ESP32, C++, Servo Control* Sep 2024 – June 2025

- Programmed ESP32 microcontroller firmware to interface with 5 servo motors and 3 sensors
- Implemented Bluetooth communication protocols for real-time control and data transmission

Autonomous VTOL Simulation | *Python, C++, Gazebo*

- Used Gazebo simulation for VTOLs and implemented hardware-in-the-loop (HITL) testing with MAVSDK
- Implemented finite state machines to enact autonomous missions, integrated with OpenCV

Frequency-Actuated Lock | *Analog Circuit Design, Op-Amps, BJT, Power Electronics*

- Designed and built analog lock that opens only at a specific audio frequency using cascaded bandpass filters, BJT amplifier stage, and op-amp gain circuits
- Implemented solenoid driver circuit with flyback diode protection to safely switch inductive loads

Inverted Pendulum Control System | *MATLAB/Simulink, Classical Control Theory, PID, OP AMP*

- Derived mathematical model of inverted pendulum system & designed stabilizing controller using PID with lead-lag compensation to achieve critically damped response
- Tuned controller gains using Ziegler-Nichols method and validated system stability through MATLAB simulation, ensuring proper settling time and minimal overshoot

Custom RISC-V Processor | *Python, SystemVerilog, Hardware Validation*

- Designed & implemented RISC-V core with modern architecture components including **caches, pipelined instructions, and a 4 stage branch predictor**
- Simulated & verified on Xilinx FPGA, implementing firmware testing functions on FPGA board via RV32
- Read and understood ISA to follow RISC-V instruction set completely, creating **SystemVerilog testbenches to complete functional coverage**

Low Power Random Number Generator accelerator | *Python, SystemVerilog, Hardware Validation*

- Implemented low power high throughput cryptography accelerator based on research paper, including 3 non linear shift registers & reset / seeding capabilities
- Created and simulated PUF (Physically unclonable function) to interface for true RNG bit generation
- Includes RAM macros for storage and transmission, carries out efficient communication via AXI protocol

Dual port DRAM accelerator | *Python, SystemVerilog, Hardware Validation*

- Designed, tested, and validated a dual port RAM module following wishbone protocol, passing gate level testing and validated for clock timing on every corner
- Includes two DRAM macros placed and validated for collision detection and high throughput, testbenches and verified through Python and bash scripting
- Designed with tradeoff between density and collision occurrences to allow for smooth use in any module

Asynchronous FIFO | *Python, SystemVerilog, Hardware Validation*

- Implemented an asynchronous FIFO to combat information transfer across clock domains, **resistant against metastability and max delays**
- Includes gray encoding schemes, addresses pointer latency, and multi bit synchronization through flip flop chains
- Tested and validated for gate level testing using open source tools like Yosys for behavioral to logical synthesis

Hardware CTF challenge | *Arduino, C, I2C*

- Transformed an arduino UNO by
-
-

MiniGPT | *PyTorch, Google Colab*

- Implemented a minimal GPT-style transformer from scratch in PyTorch to understand core LLM architecture
- Built essential components including tokenization, multi-head self-attention, and positional embeddings
- Trained and evaluated the model on small text data to validate end-to-end sequence generation

MNIST ANN | *CUDA, Google Colab, C/C++*

- Built and bench marked matmul kernel from scratch reaching 80% speed of default matmul
- Utilized shared memory, caching optimizations, and other CUDA techniques to speed up inference by 120%
- Implemented convolutional kernels using IM2COL & optmized matmul kernels

CUDA PTX matmul | *CUDA, PTX*

- Wrote matmul kernel in PTX
-
-

Technical Skills

Programming Languages/ Methodologies: C / C++, Python, RV32 Assembly, Verilog / VHDL, UVM
Embedded Systems: Bare-metal Firmware, FreeRTOS, Device Drivers, Hardware Bring-up
Protocols: PCIE, SATA, AXI, Wishbone, I2C, SPI, UART,
Development Tools: Oscilloscope, Logic Analyzer,CMake, Git, Docker, Perl, Shell, Bash, Yosys, Openlane, UVM